Local Differential Privacy: Refined Mechanism Design and Utility Analysis

Ye Zheng

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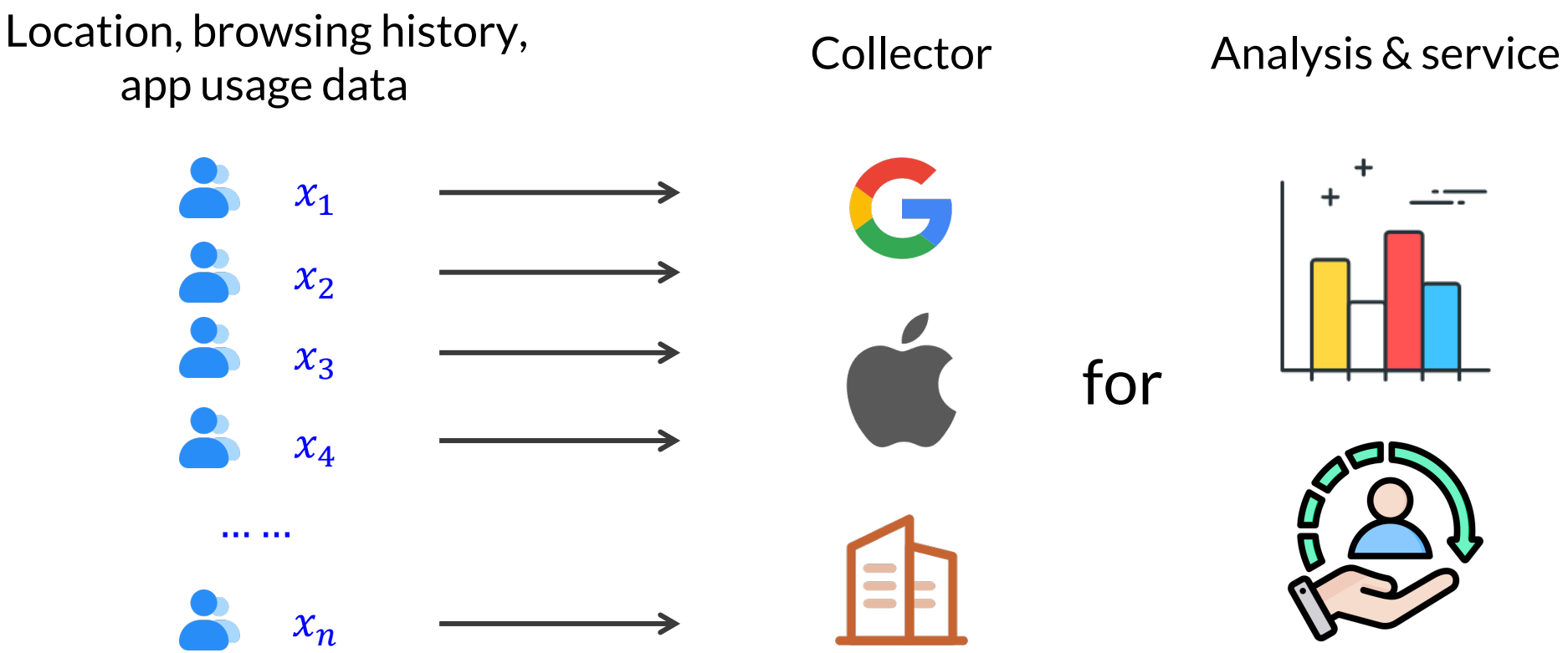
RIT | Rochester Institute of Technology

PDF & slides 👉 https://zhengyeah.com

Data Collection Everywhere

RIT

- Users' personal data are collected by companies for analysis or services



Users' Data Privacy

RIT

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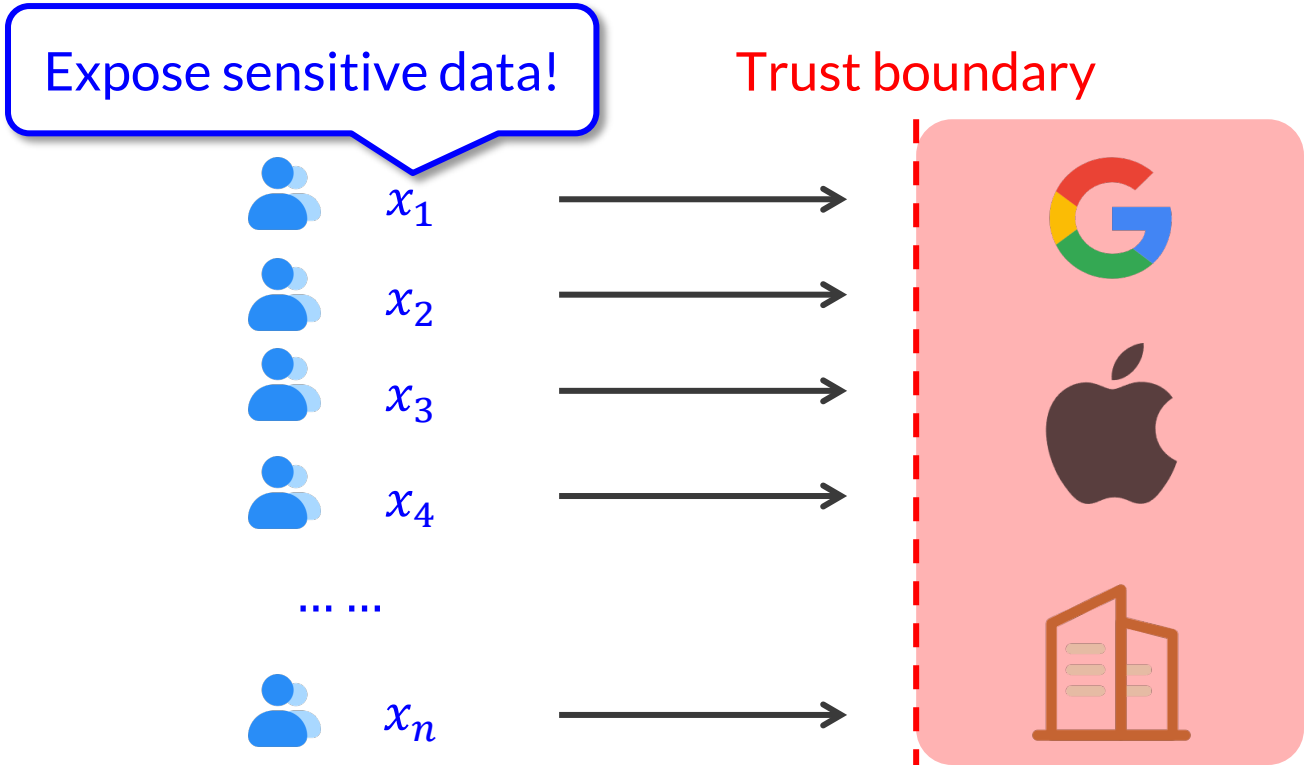
  - these companies may not be trusted to collect users' sensitive data



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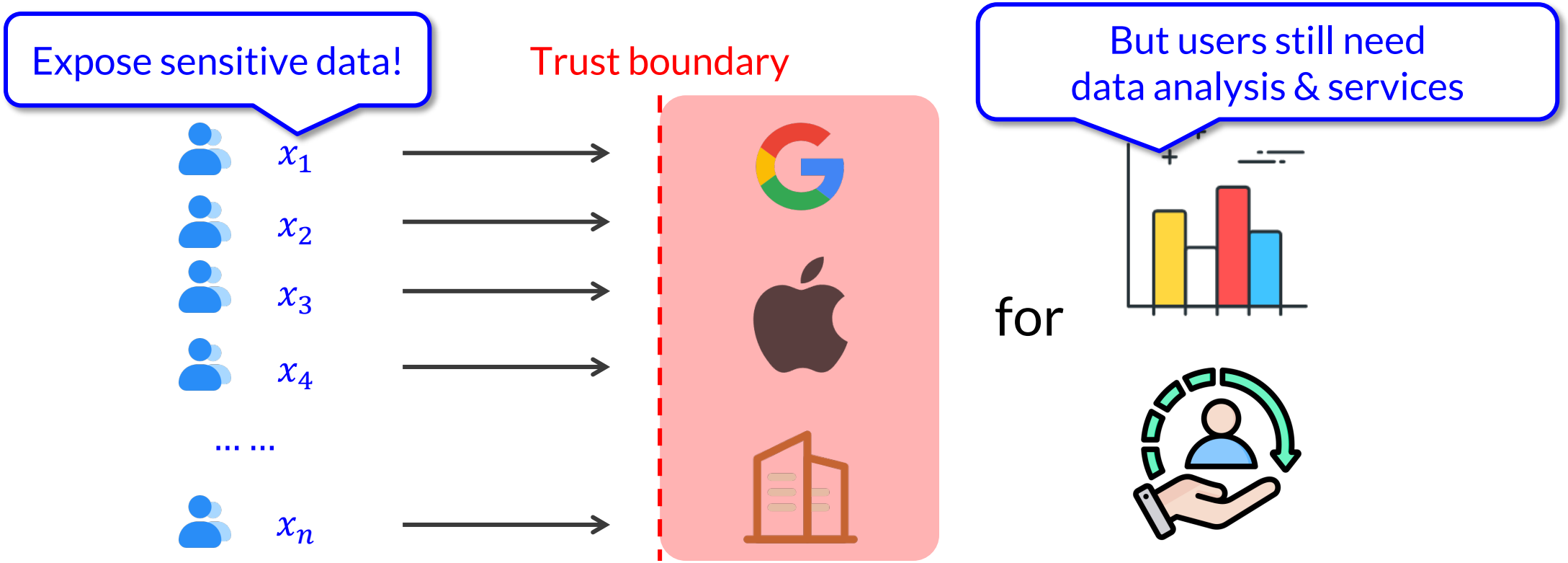

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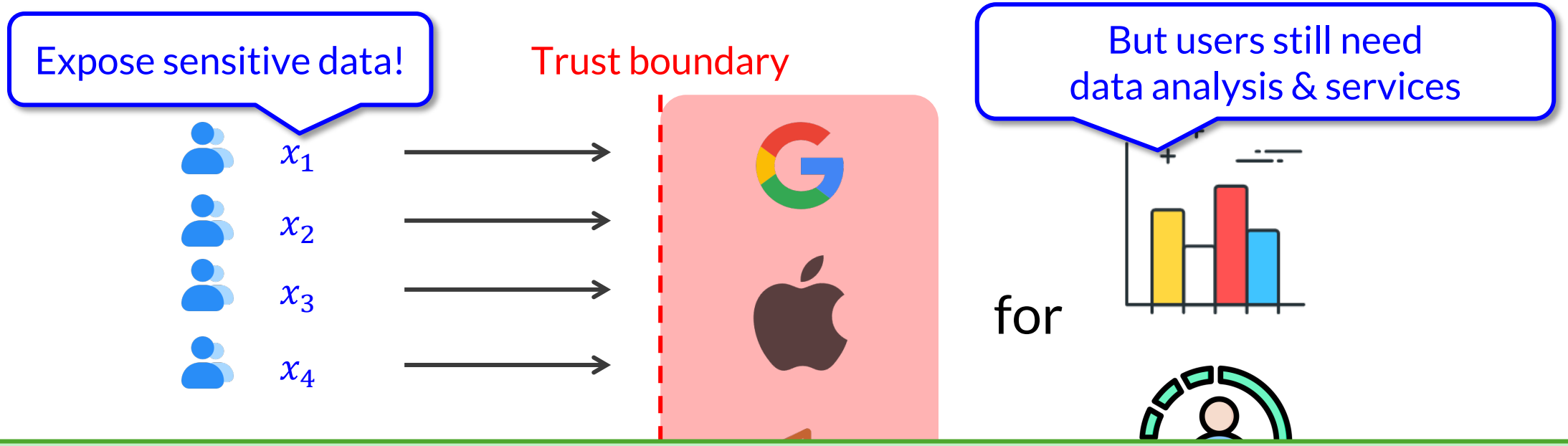
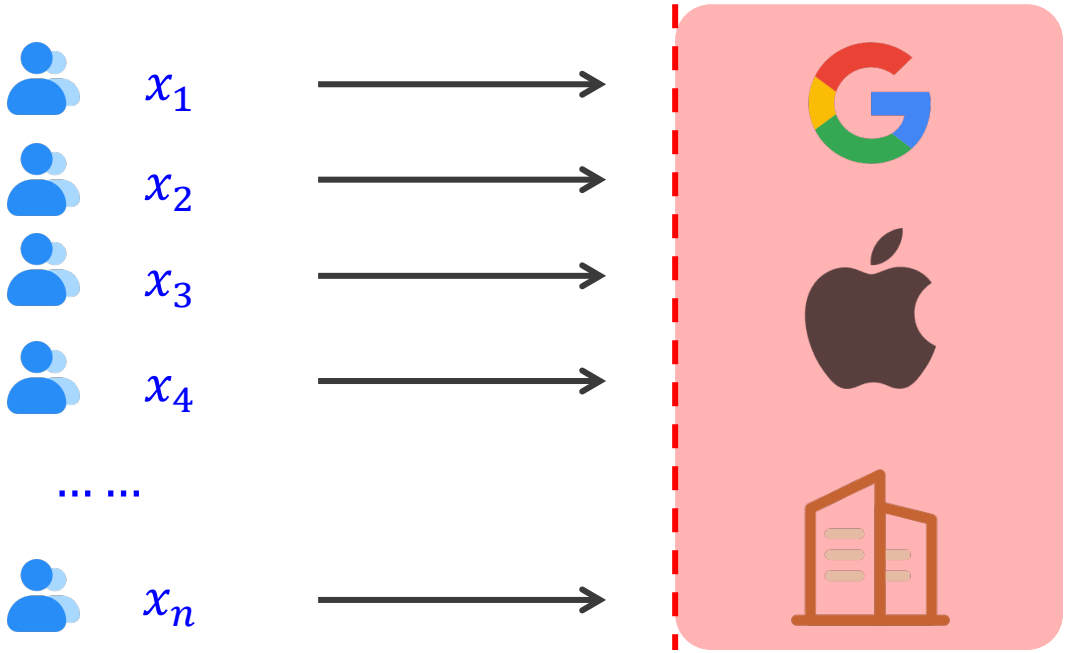


Q: How can we provide data analysis & services **while** protecting users' data privacy?

Privacy-Preserving Computation

RIT

- Users' personal data are collected by companies for analysis or services
  - these companies may be untrusted to collect users' sensitive data
  - how to compute $f(x_1, x_2, \dots, x_n)$ without revealing $x_1, x_2, \dots, x_n$?



for $f(x_1, x_2, \dots, x_n)$

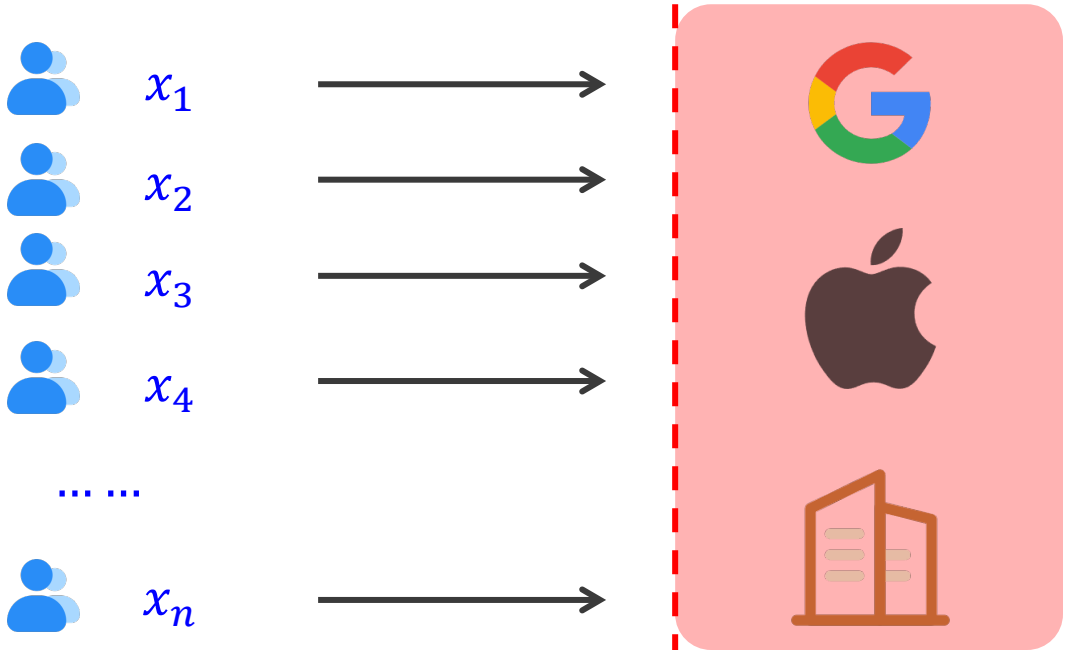
Privacy-Preserving Computation - **Techniques**

- Homomorphic encryption (HE), multi-party computation (MPC), local differential privacy (LDP), etc

Privacy-Preserving Computation - **HE**

RIT

- Homomorphic encryption (HE):
  - "homomorphic": preserving structure
  - design algorithms $\{Enc, Dec\} \rightarrow Dec\big(f(Enc(x_1), Enc(x_2), \dots, Enc(x_n))\big) = f(x_1, x_2, \dots, x_n)$



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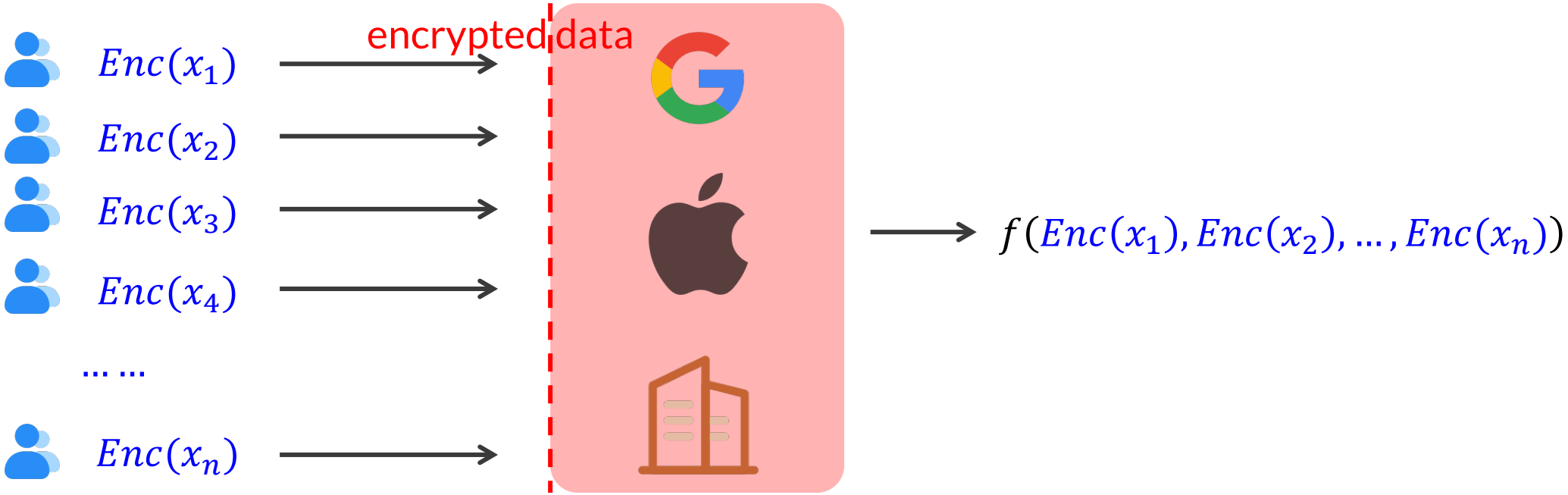

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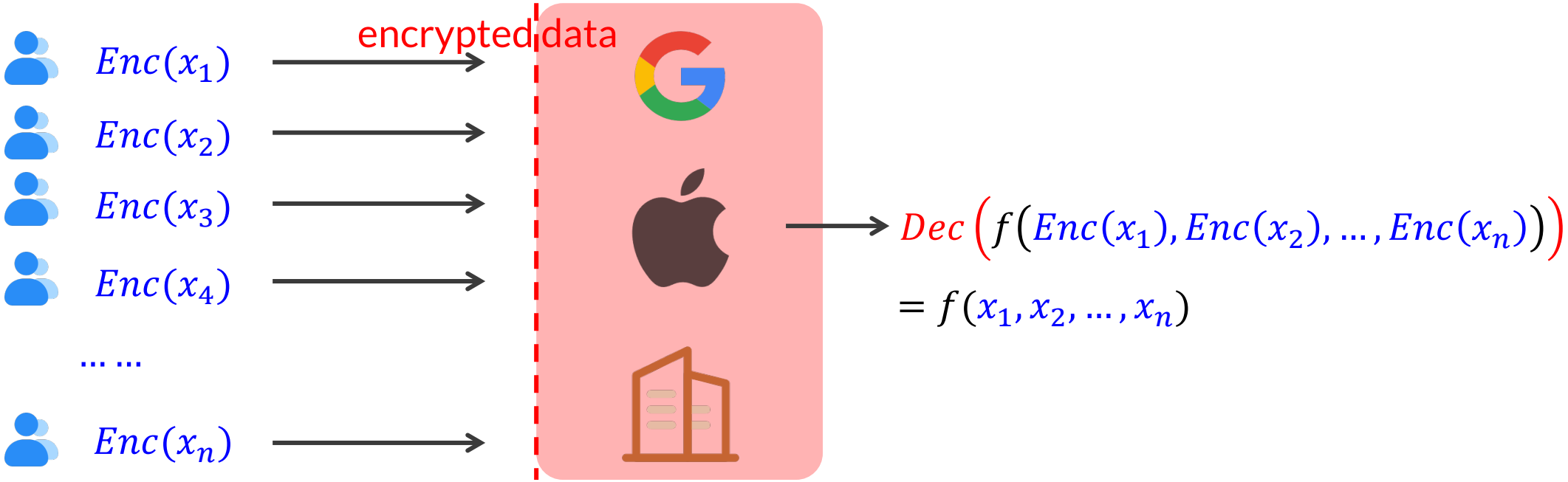

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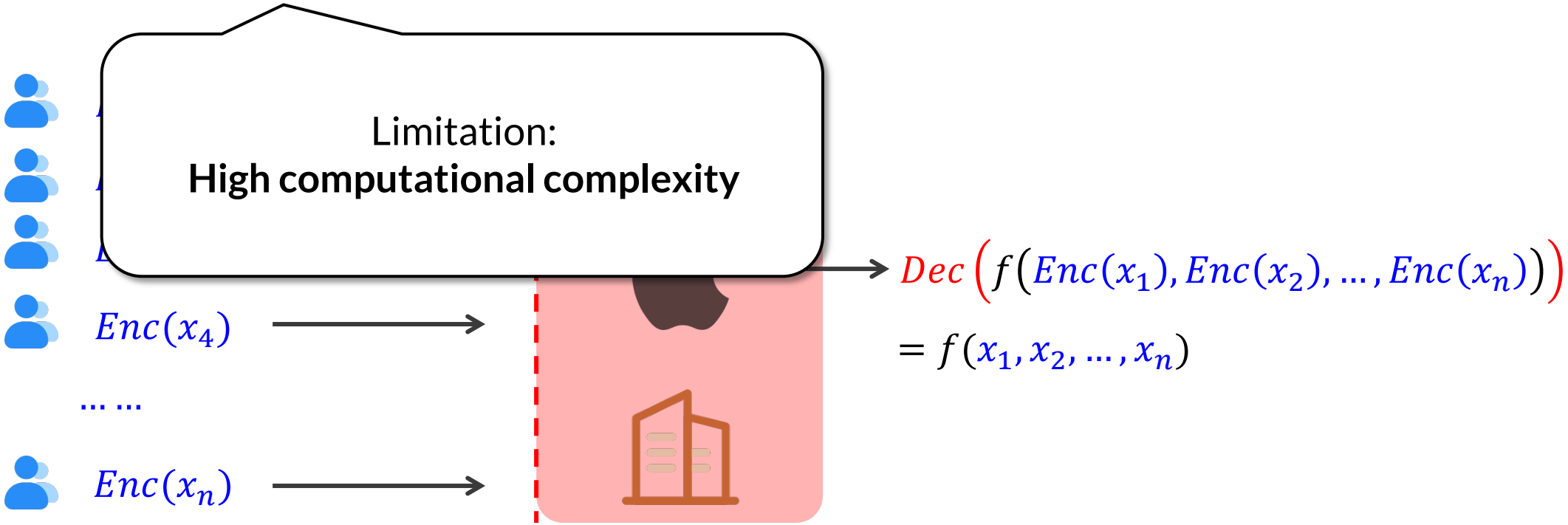

Privacy-Preserving Computation - **MPC**

RIT

- Multi-party computation (MPC):
  - no central party
  - jointly compute f without revealing x_i
- Example: $f(x_1, x_2) = x_1 + x_2$

x_1

for $f(x_1, x_2)$

x_2

… …

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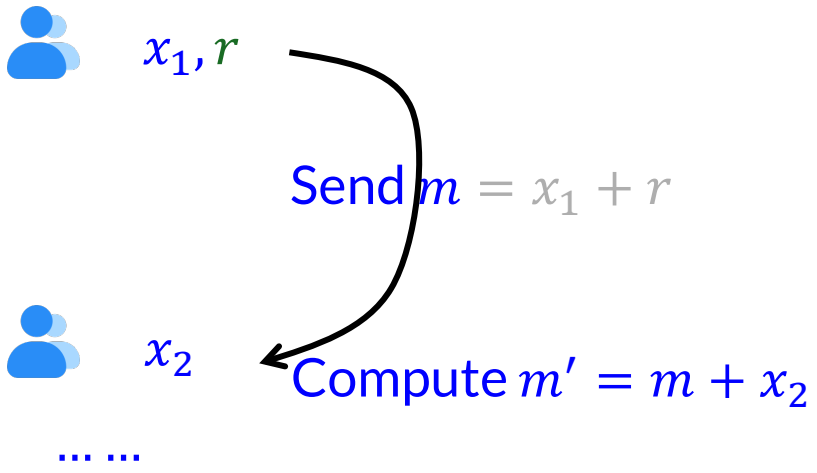

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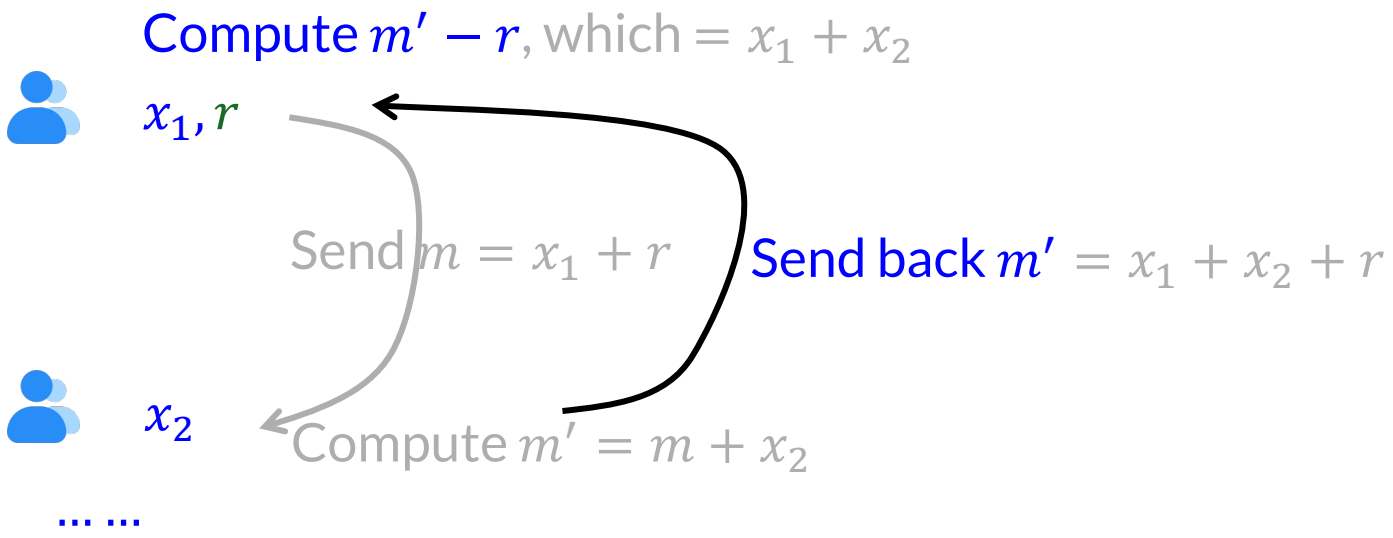

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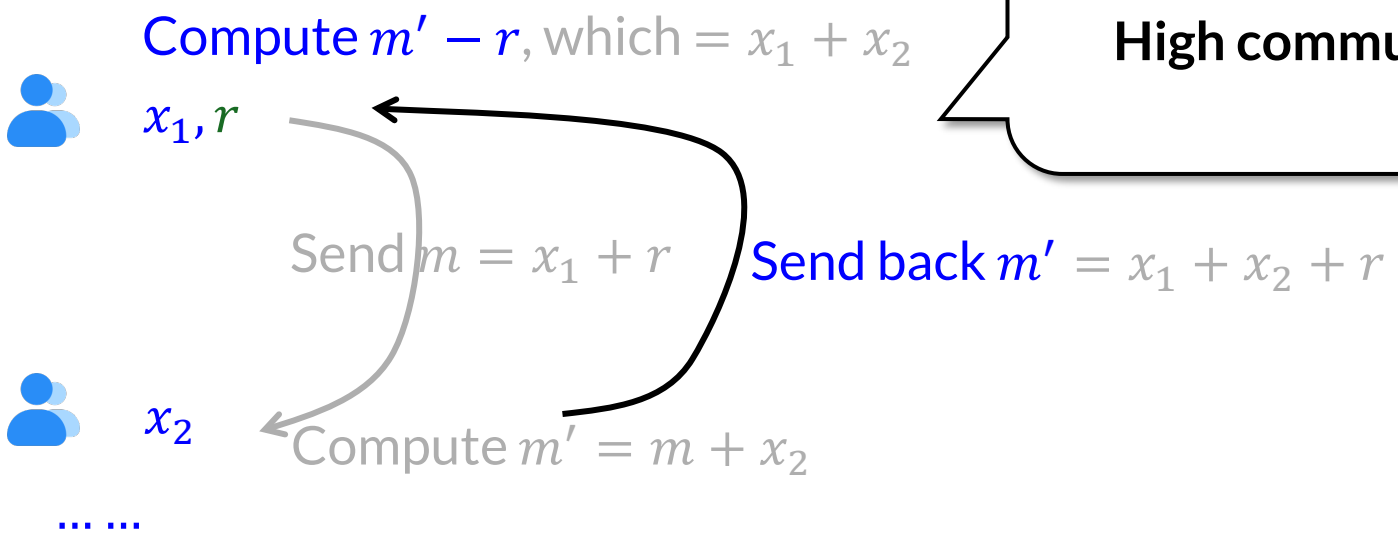

Limitation:
High communication complexity

Privacy-Preserving Computation - **LDP**

RIT

- Local differential privacy (LDP):
  - hard to differentiate the sensitive data from other data
  - each user locally perturbs x_i to $\tilde{x}_i$ $\rightarrow$ $f(\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n) \approx f(x_1, x_2, \dots, x_n)$

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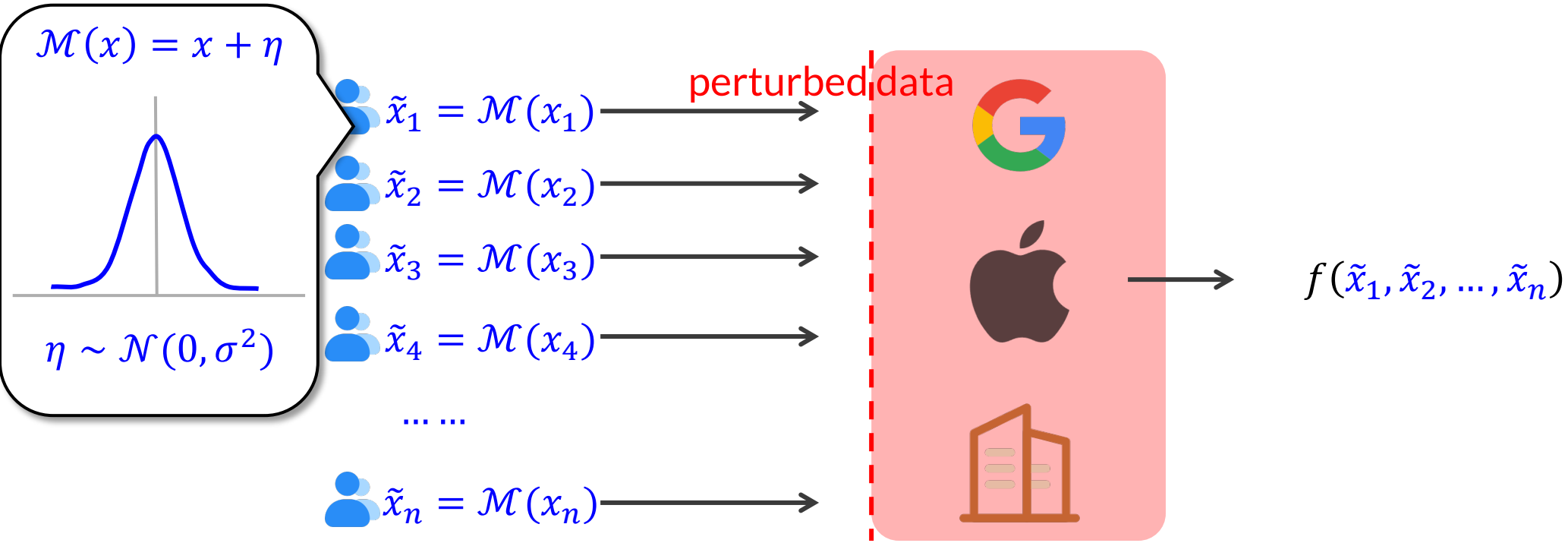

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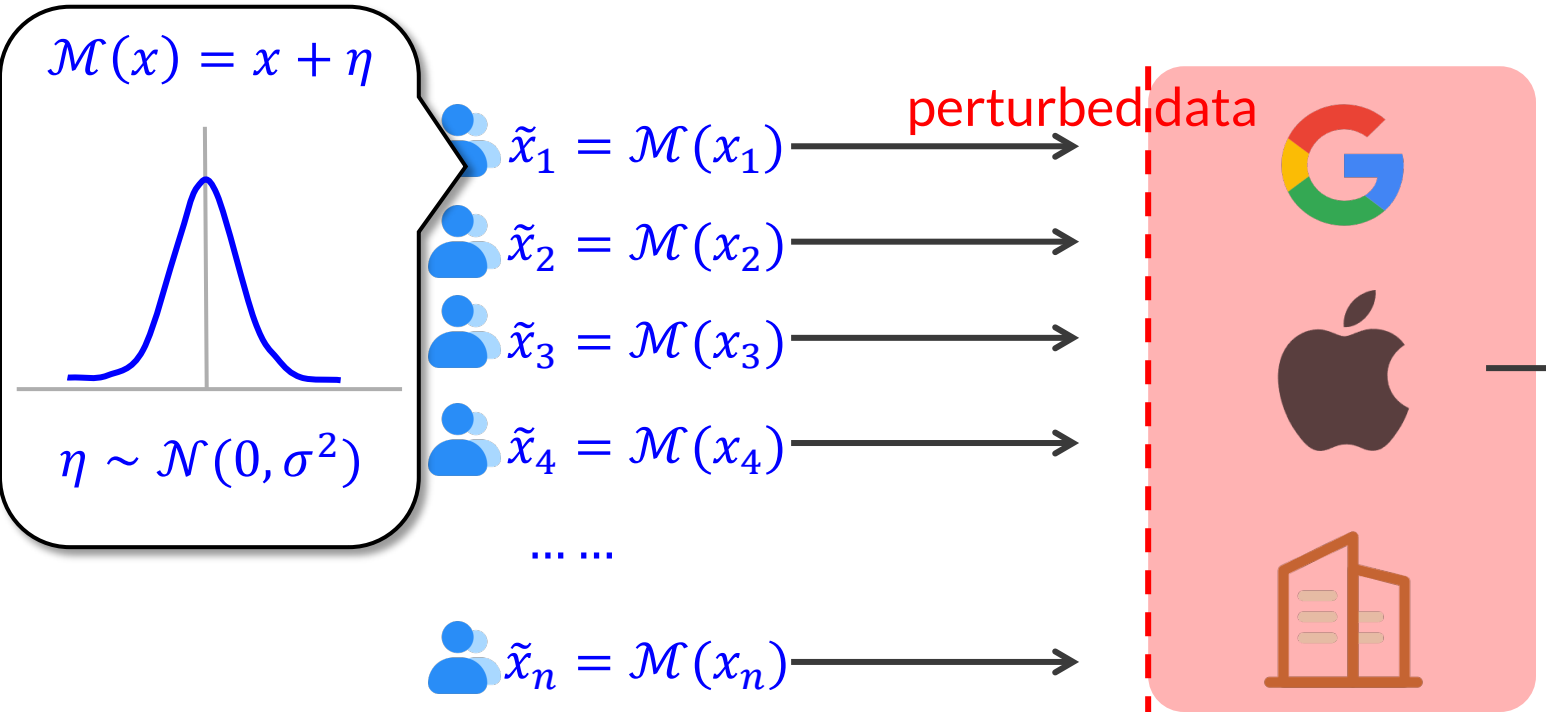

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Privacy-Preserving Computation - **LDP**

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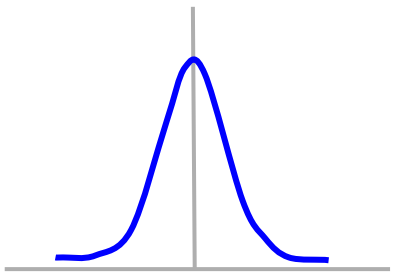
Advantages:
Negligible computational complexity
No communication between users

But approximated f

$\mathcal{M}(x) = x + \eta$



$\eta \sim \mathcal{N}(0, \sigma^2)$

$\tilde{x}_1 = \mathcal{M}(x_1)$

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… …

$\tilde{x}_n = \mathcal{M}(x_n)$

perturbed data

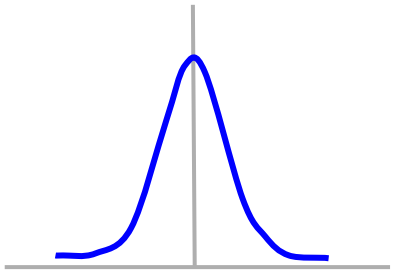
$f(\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n)$

$\mathrm{E}[f(\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n)] = f(x_1, x_2, \dots, x_n)$

LDP – Applications

RIT

Chrome uses LDP to collect homepage settings, extension usage, etc



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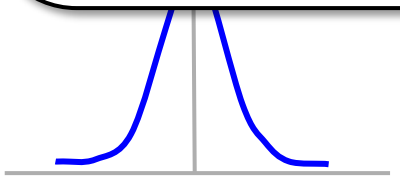
Emoji usage, new keyboard words, Safari URL statistics, health analytics



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LDP – Formal Privacy

RIT

- After applying $\mathcal{M}$, the confidence of distinguishing sensitive x_1 and x_2 from observation $\tilde{x}$:

$$\forall x_1, x_2 \in \mathcal{D}, \forall y \in \widetilde{\mathcal{D}} \qquad \max \frac{\Pr[\mathcal{M}(x_1) = \tilde{x}]}{\Pr[\mathcal{M}(x_2) = \tilde{x}]} \leq e^{\varepsilon}$$

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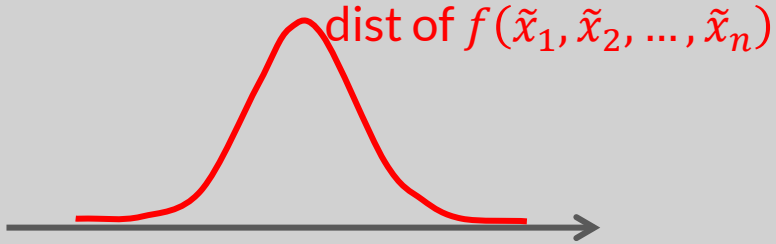
- The collector's / adversary's view: hard to infer the sensitive data

Privacy quantified by ε

$x_1 \rightarrow \mathcal{M} \rightarrow \tilde{x}$

Provable defense against data inference attacks

Data utility by approximated error



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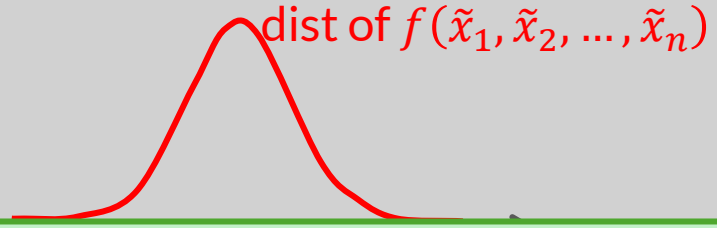

Fundamental direction: **Design** of $\mathcal{M}$ to **optimize** the privacy—utility tradeoff

LDP – Formal Privacy

RIT

Utility analysis of $f \circ \mathcal{M}$

$$f(x_1, x_2, \dots, x_n) := \sum_{i=1}^{n} x_i \quad \text{or} \quad f(x_1, x_2, \dots, x_n) := \{x_1, x_2, \dots, x_n\} \rightarrow \text{Variance, MSE}$$

$$f(x_1, x_2, \dots, x_n) := h: \mathbb{R}^n \rightarrow \{1,2, \dots, K\} \text{ is a classifier} \rightarrow$$

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Data utility by app… error

dist of $f(\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n)$

Fundamental direction: **Utility analysis** of complex task f

This Proposal: LDP Theory

RIT

- Advancing LDP's mechanism design and utility analysis

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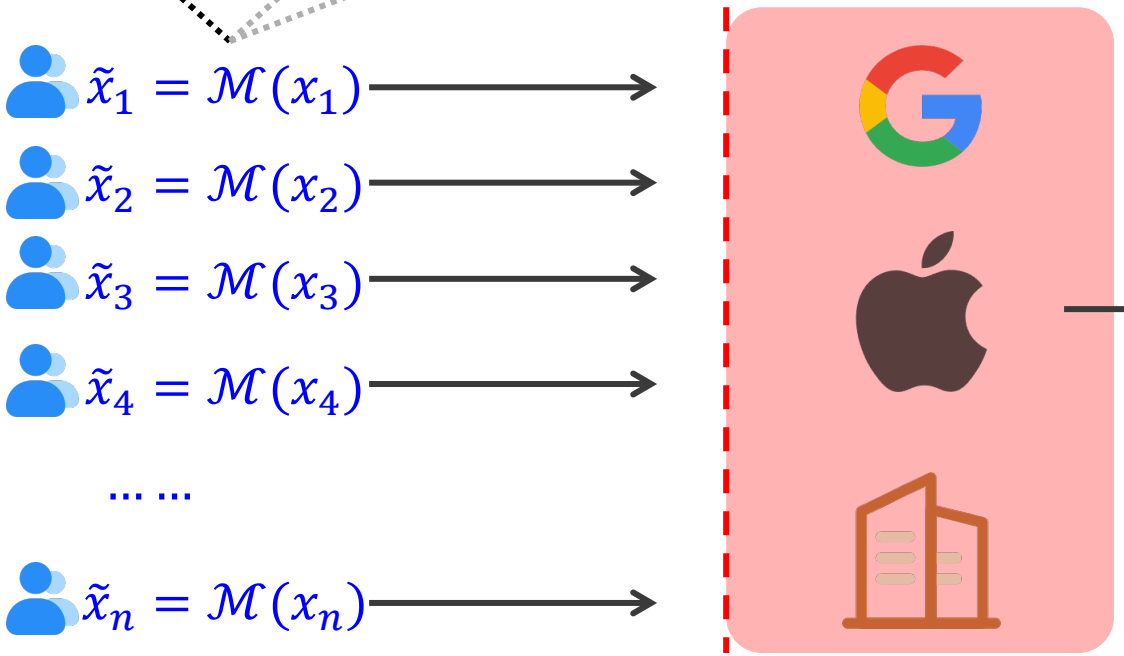
RIT

- Advancing LDP's mechanism design and utility analysis

Part 1: correlated $\mathcal{M}$

Part 2: optimal piecewise-based $\mathcal{M}$

Part 3: $\mathcal{M}$ for trajectories in continuous space



Part 4: utility analysis for classifier $\circ\ \mathcal{M}$

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This Proposal: LDP Theory

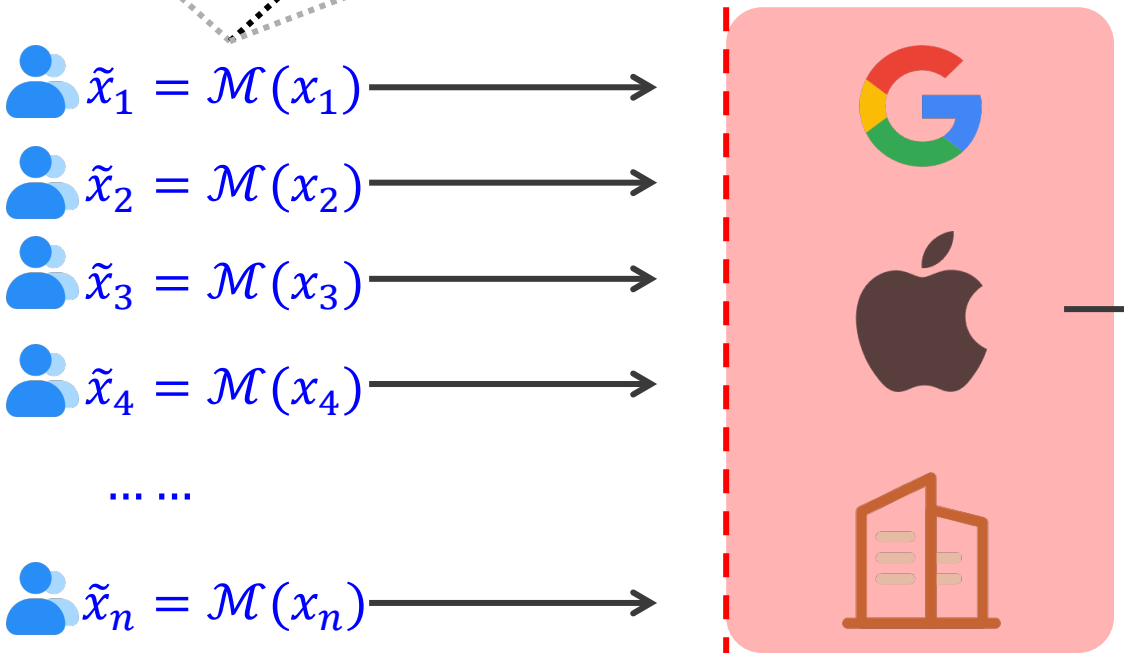
RIT

- Advancing LDP's mechanism design and utility analysis

binary x → numerical x

Part 1: correlated $\mathcal{M}$ → Part 2: optimal piecewise-based $\mathcal{M}$

Part 3: $\mathcal{M}$ for trajectories in continuous space



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f is a classifier

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1D x → 2D x

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Mechanism-level
↓
Task-level

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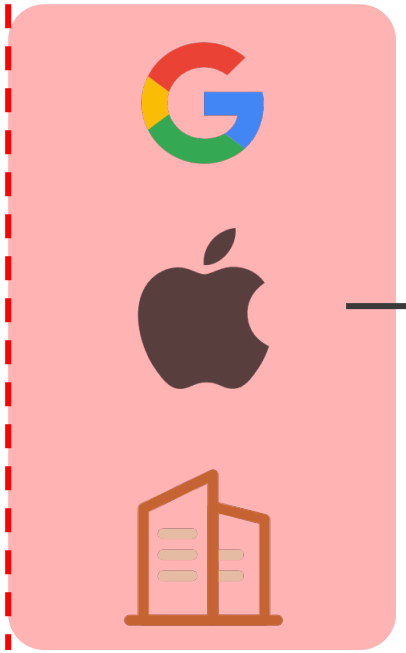

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Part 1 – Correlated $\mathcal{M}^*$

RIT

- Existing LDP mechanisms: Each user perturbs their data independently

$x_i \longrightarrow \mathcal{M}(x_i) \longrightarrow$

$x_j \longrightarrow \mathcal{M}(x_j) \longrightarrow$

* [PETS'25] Locally Differentially Private Frequency Estimation via Joint Randomized Response

Part 1 – Correlated $\mathcal{M}^*$

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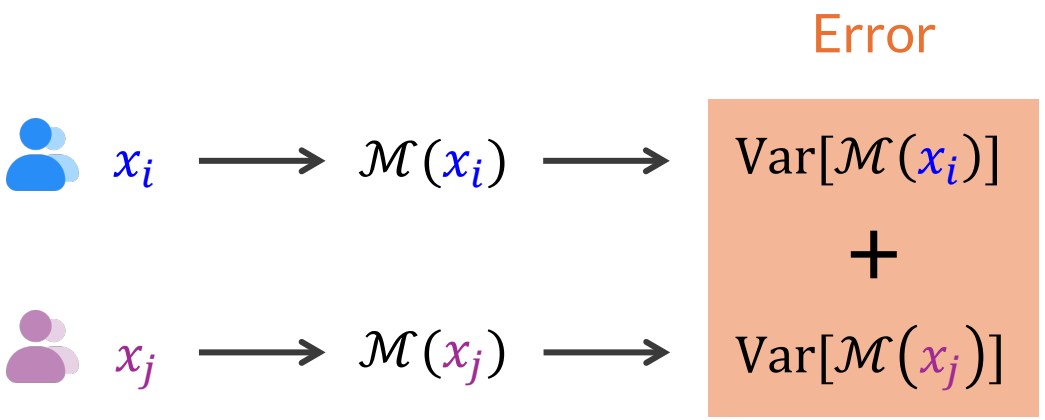

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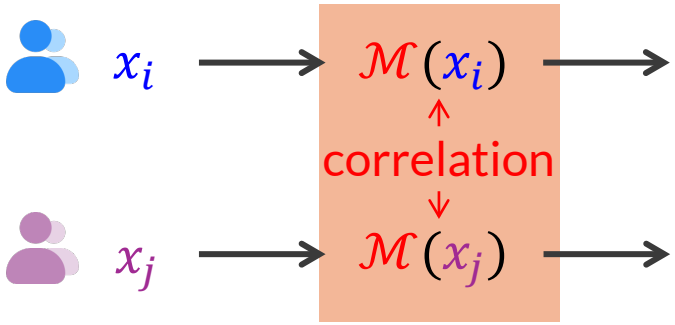
- Correlated LDP mechanisms: Users' data are perturbed by correlated $\mathcal{M}$



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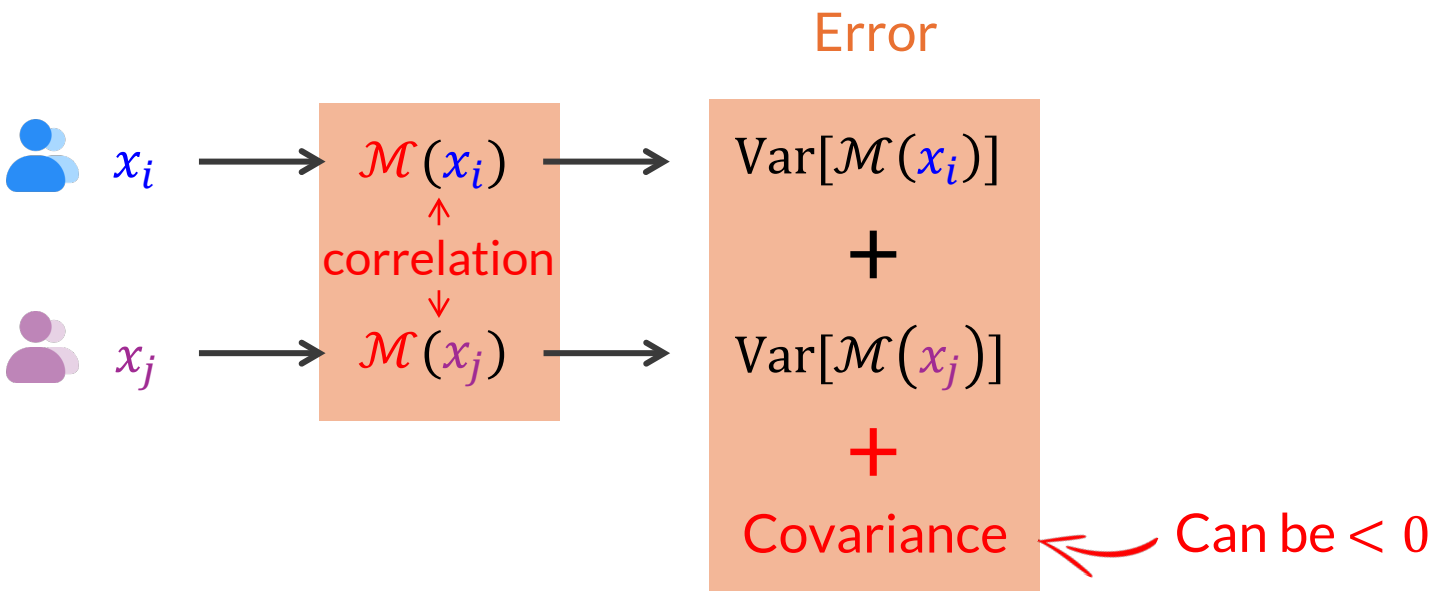

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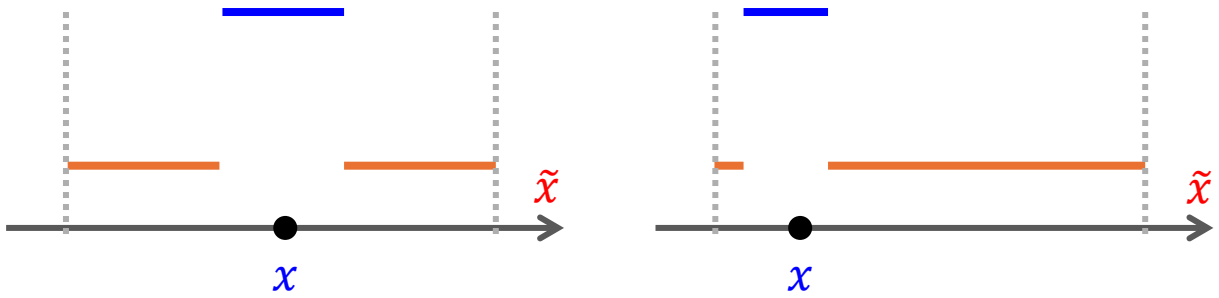
Part 2 – Optimal Piecewise-based $\mathcal{M}^{\dagger}$

RIT

binary x → numerical x

- SOTA for bounded numerical data: Piecewise-based mechanisms (3-piece heuristic PDF)

$\tilde{x} \leftarrow \mathcal{M}(x)$: sampling PDFs



† [PETS'25] Optimal Piecewise-based Mechanism for Collecting Bounded Numerical Data under Local Differential Privacy

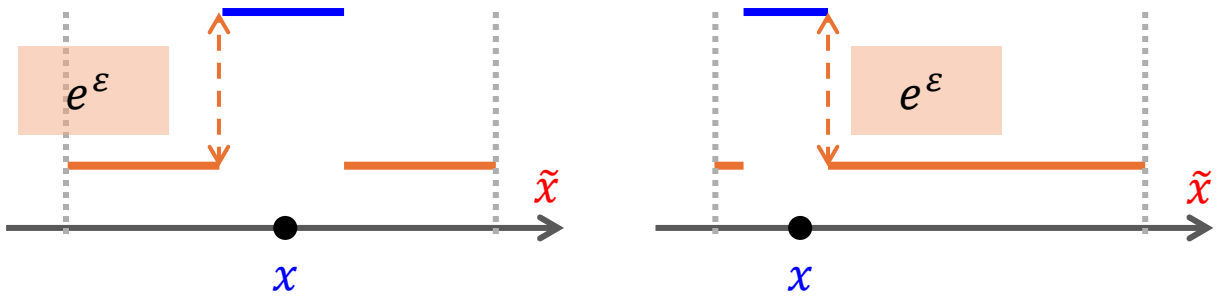
Part 2 – Optimal Piecewise-based $\mathcal{M}^{\dagger}$

RIT

binary x → numerical x

- SOTA for bounded numerical data: Piecewise-based mechanisms (3-piece heuristic PDF)

$\tilde{x} \leftarrow \mathcal{M}(x)$: sampling PDFs



$$\mathrm{pdf}[\mathcal{M}(x) = \tilde{x}] = \begin{cases} p_{\varepsilon} & \text{if } \tilde{x} \in [l_{x,\varepsilon}, r_{x,\varepsilon}] \\ \dfrac{p_{\varepsilon}}{e^{\varepsilon}} & \text{if } \tilde{x} \in \tilde{\mathcal{D}} \backslash [l_{x,\varepsilon}, r_{x,\varepsilon}] \end{cases}$$

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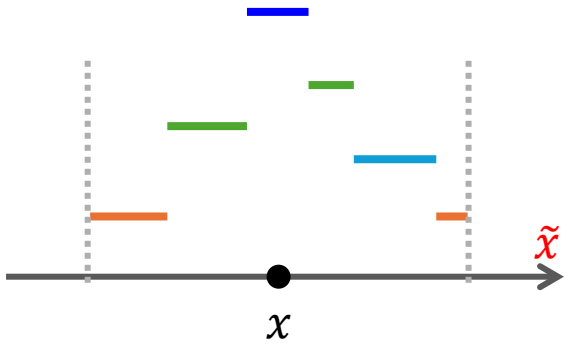

Potentially has lower error

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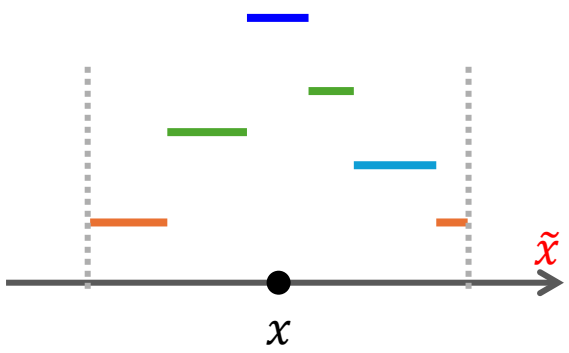


Potentially has lower error

- What is the optimal piecewise-based mechanism?

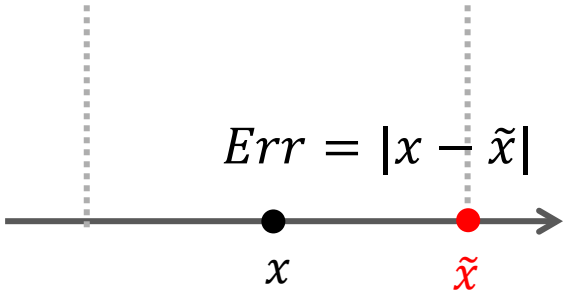
† [PETS'25] Optimal Piecewise-based Mechanism for Collecting Bounded Numerical Data under Local Differential Privacy

Part 2 – Optimal Piecewise-based $\mathcal{M}$ †

RIT

binary x → numerical x

- Linear data domain → Circular data domain



† [PETS'25] Optimal Piecewise-based Mechanism for Collecting Bounded Numerical Data under Local Differential Privacy

Part 2 – Optimal Piecewise-based $\mathcal{M}$†

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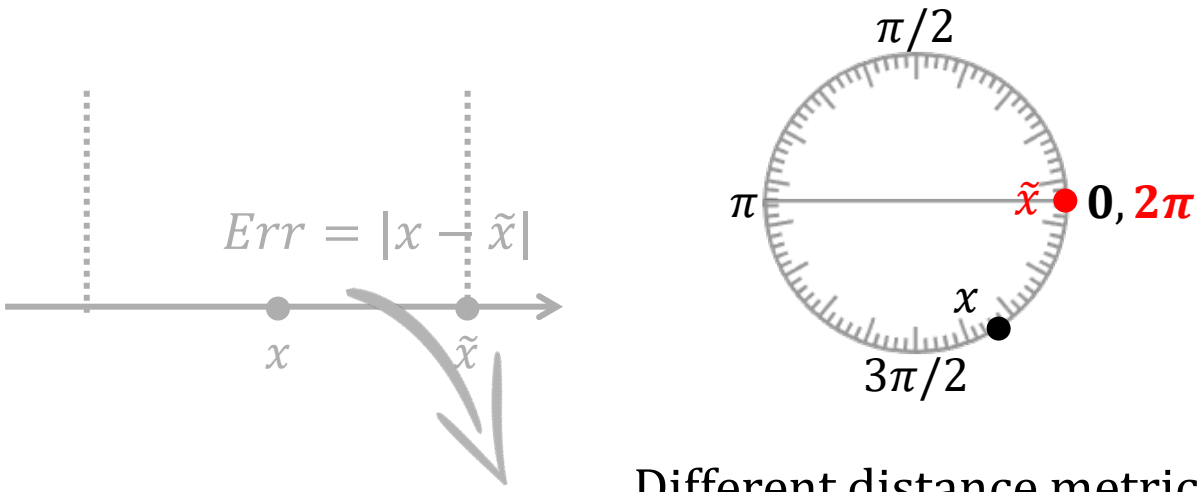

Different distance metric

$$Err = \min(|x - \tilde{x}|, |2\pi - x - \tilde{x}|)$$

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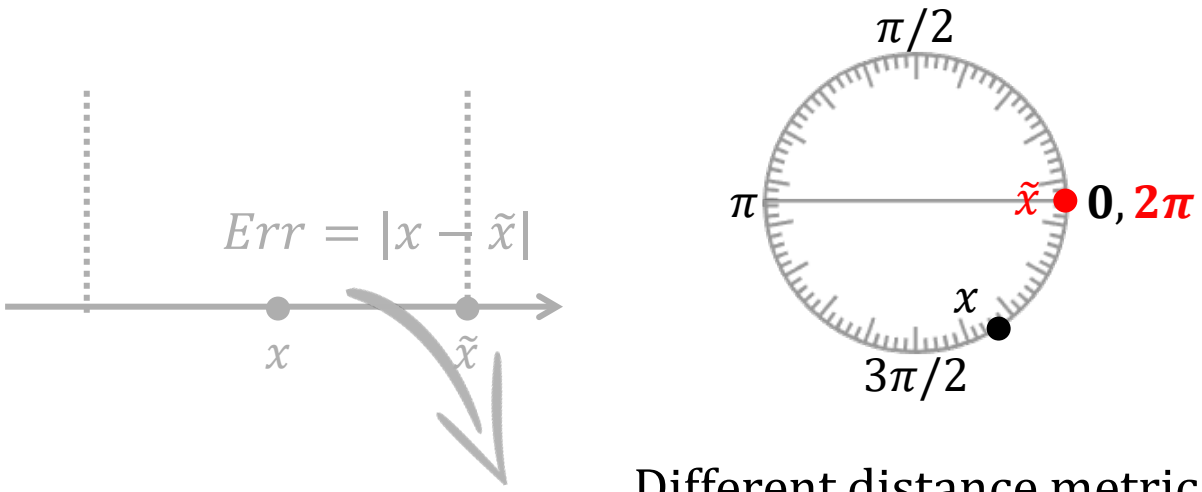


Different distance metric

$$Err = \min(|x - \tilde{x}|, |2\pi - x - \tilde{x}|)$$

- What is the optimal piecewise-based mechanism for circular domain?

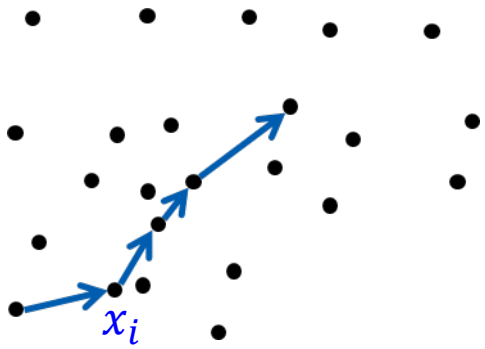
† [PETS'25] Optimal Piecewise-based Mechanism for Collecting Bounded Numerical Data under Local Differential Privacy

Part 3 – $\mathcal{M}$ for Trajectories in Continuous Space‡

1D x → 2D x

- Existing $\mathcal{M}$ for trajectory collection: assuming discrete location space

$$\mathcal{S} = \{p_1, \dots, p_n\}$$



‡ [PETS'26] TraCS: Trajectory Collection in Continuous Space under Local Differential Privacy

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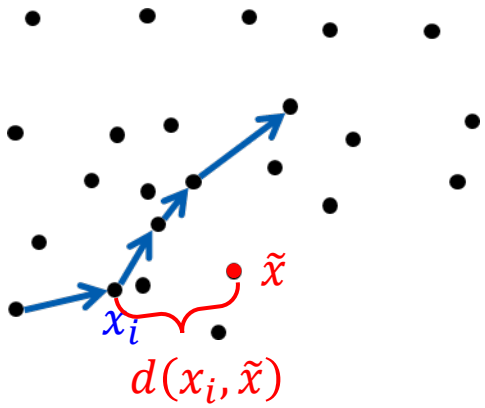

Make each x_i satisfying LDP

For $\tilde{x} \in S$

$$\Pr[\mathcal{M}(x_i) = \tilde{x}] \propto e^{d(x_i, \tilde{x})}$$

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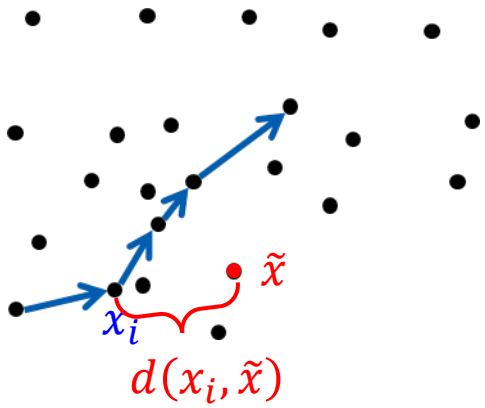

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Limitations

- expensive to sample
- only applicable to discrete S

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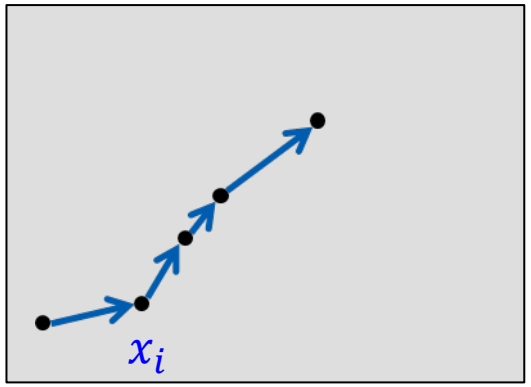
Part 3 – $\mathcal{M}$ for Trajectories in Continuous Space‡

1D x → 2D x

- Discrete location space → Continuous location space

$$\mathcal{S}_{\text{dis}} \subset \mathcal{S} = [0,1.5] \times [0,1]$$



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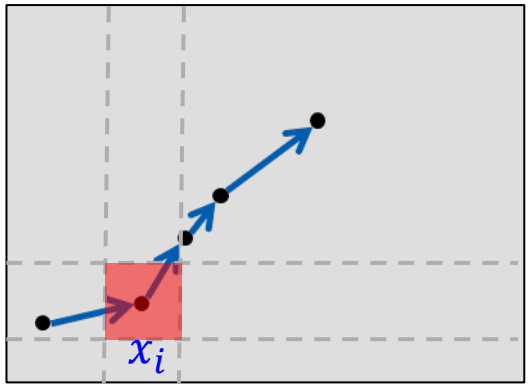

Make each x_i satisfying LDP in $\mathcal{S}$

$\Pr[\mathcal{M}(x_i) = \tilde{x} \in$ ■ $] = p_\varepsilon$

$\Pr[\mathcal{M}(x_i) = \tilde{x} \in$ ■ $] = p_\varepsilon / e^\varepsilon$

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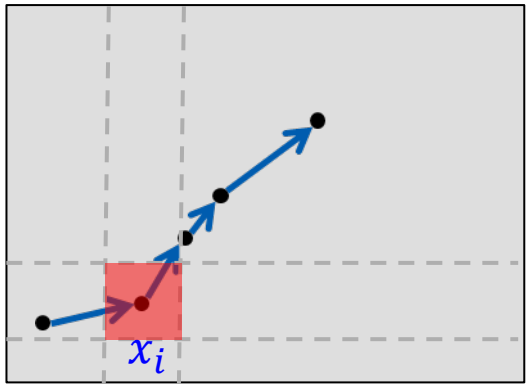


Make each x_i satisfying LD

$\Pr[\mathcal{M}(x_i) = \tilde{x} \in$] $= p_\varepsilon$

$\Pr[\mathcal{M}(x_i) = \tilde{x} \in$] $= p_\varepsilon / e^{\varepsilon}$

Benefits

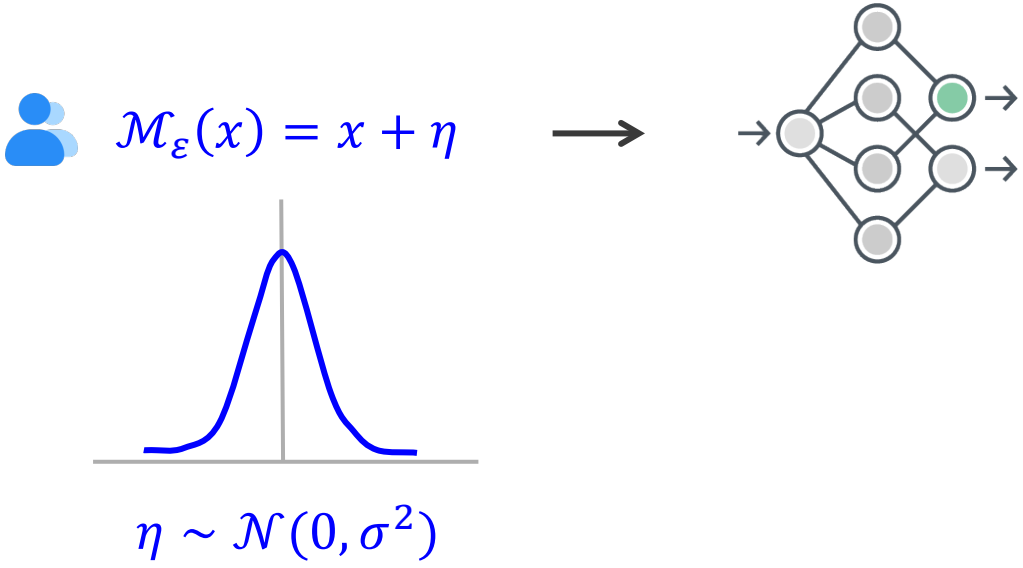
- negligible sampling complexity
- applicable to discrete space by post-processing

‡ [PETS'26] TraCS: Trajectory Collection in Continuous Space under Local Differential Privacy

Part 4 – Utility Analysis for classifier $\circ\ \mathcal{M}$¶

mechanism-level → task level

- Empirical classifier utility under $\mathcal{M}$



¶ [PETS'26] Quantifying Classifier Utility under Local Differential Privacy

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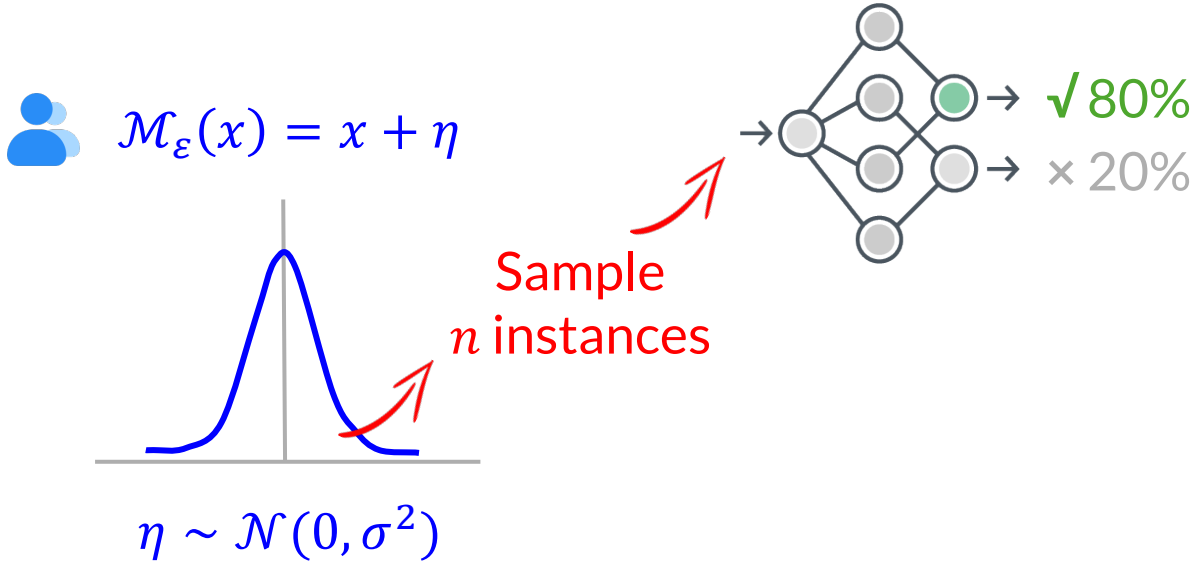

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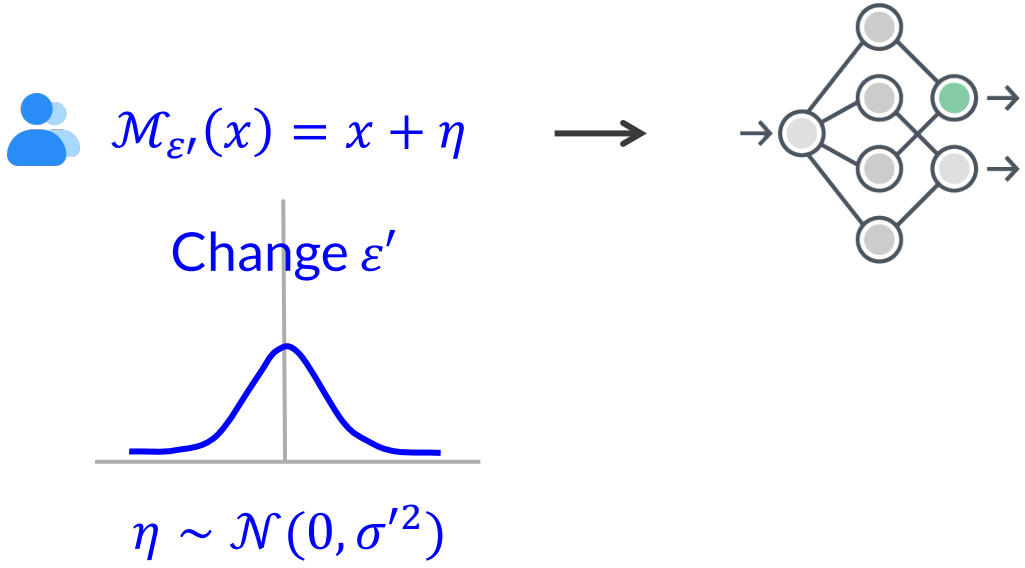

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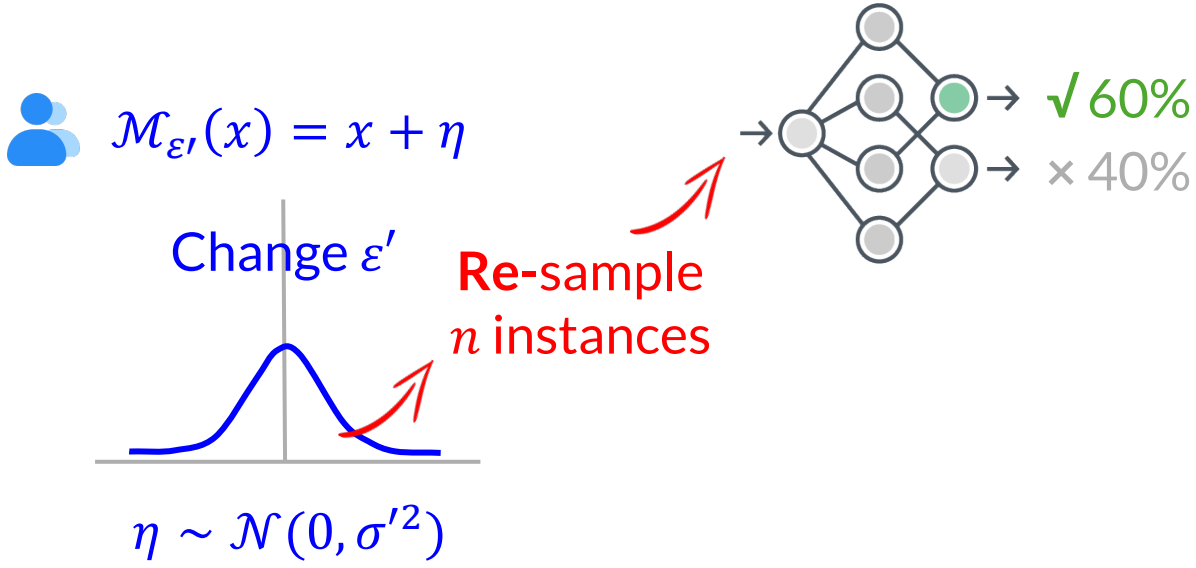

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Part 4 – Utility Analysis for classifier $\circ\ \mathcal{M}$¶

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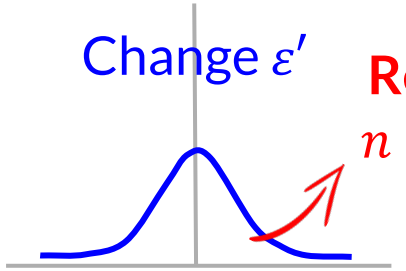
- Empirical classifier utility under $\mathcal{M}$

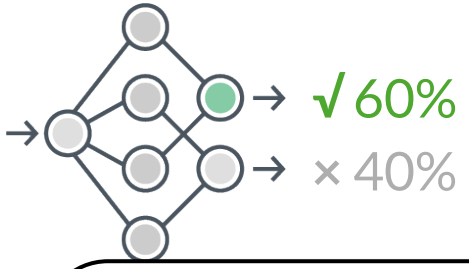
$\mathcal{M}_{\varepsilon'}(x) = x + \eta$

Change ε'



$\eta \sim \mathcal{N}(0, \sigma'^2)$

Re-sample n instances



Limitation

Empirical result
&
time-consuming resampling

¶ [PETS'26] Quantifying Classifier Utility under Local Differential Privacy

Part 4 – Utility Analysis for classifier $\circ\ \mathcal{M}$¶

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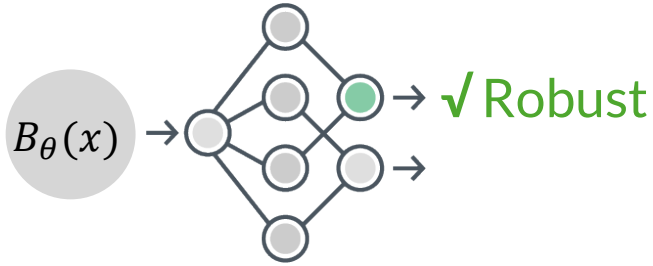
- Empirical classifier utility under $\mathcal{M}$ → Analytical classifier utility



Robustness:
$B_\theta(x)$ is robust

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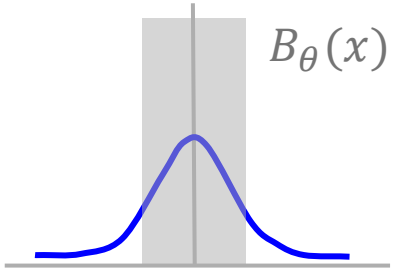

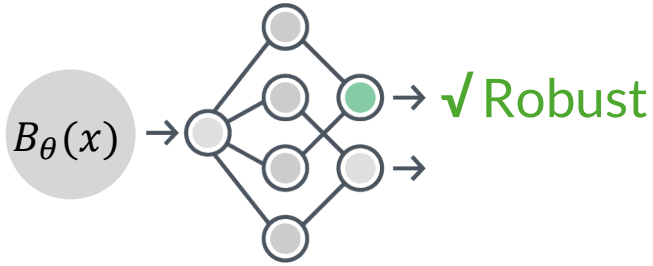

Concentration:
$\Pr[\mathcal{M}_\varepsilon(x) \in B_\theta(x)]$
$:= p(\varepsilon, \theta)$

Robustness:
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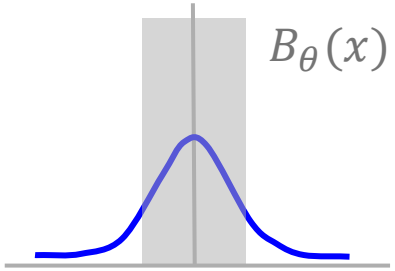

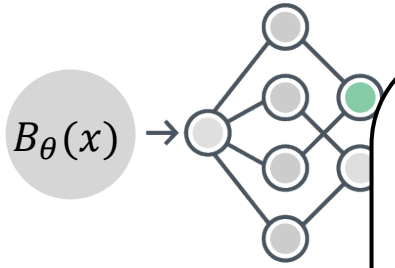

Concentration:
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$:= p(\varepsilon, \theta)$

Robustn
$B_\theta(x)$ is

Connected by θ

Analytical classifier utility

With probability at least $p(\varepsilon, \theta)$, the classifier preserves its correct classification under input $\mathcal{M}_\varepsilon(x)$.

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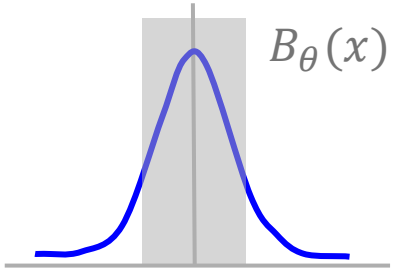

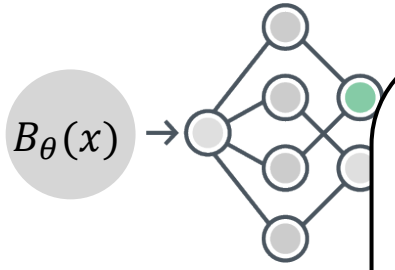

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¶ [PETS'26] Quantifying Classifier Utility under Local Differential Privacy

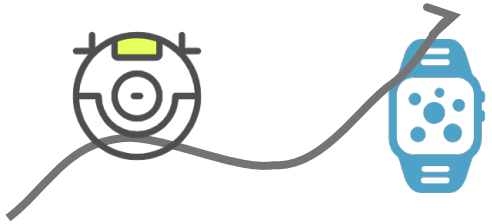
Social Contributions

RIT

- **New LDP building blocks**
  - correlated LDP mechanisms
  - optimal piecewise-based mechanisms

Sensor networks & Federated learning, etc

- **Universal trajectory collection mechanisms**
  - applicable to both continuous / discrete space



Smart home & wearable devices' trajectories, etc

- **Analytical view of classifier utility under LDP-perturbed inputs**
  - choosing ε for when using classifiers